

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2025-2026

Mid-Semester Test
(EC-2 Regular)

Course No. : AIMLZ565
Course Title : Machine Learning
Nature of Exam : Closed Book
Weightage : 30%
Duration : 2 Hours
Date of Exam : 15/03/2026 (FN/AN)

No. of Pages	= 4
No. of Questions	= 6

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Please Note:

A **maximum of 12 main questions** with Q.Nos. 1 to N. **Please do not include any objective type questions. Please do not include any reference material like data tables in the question paper, as all Mid-Semester Test are completely Closed Book examinations. Both the options to submit answers i.e. typing as well as scanning & uploading hand written answers should be available for ALL the questions.**

- Q.1 A city government wants to understand how annual sales tax collections change with the number of new car registrations each year. To support budgeting and revenue forecasting, the government collected the following historical data for 7 years, where:
X = Number of new car registrations (in thousands)

Y = Annual sales tax collections (in suitable units, e.g., million/crore)

Year	X (Registrations in thousands)	Y (Sales tax collections)
1	10	1
2	12	1.4
3	15	1.9
4	16	2
5	14	1.8
6	17	2.1
7	20	2.3

Assume a linear relationship between registrations and sales tax collection

$$\hat{Y} = \theta_0 + \theta_1 X \quad \text{where the left hand side refers to the predicted value of Y.}$$

- (A) **[4 Marks]** Using the closed-form (Normal Equation) method of Ordinary Least Squares (OLS), determine the least-squares regression equation.
- (B) **[2 Marks]** Using the fitted regression equation, estimate the annual sales tax collections when the city records 22,000 new car registrations (i.e., $X=22$).
- (C) **[4 Marks]** Consider the following training set with 5 examples and regression model as $Y=3-4X+2X^2$

X	Y
5	30
8	90
12	250
15	498
20	900

Calculate the following error measures.

- Root Mean Square Error
- Mean Absolute Error

Q.2

- (a) **[4 Marks]** A dataset contains the following attributes: Temperature ($^{\circ}\text{C}$), Salary (₹), Customer Satisfaction (Low/Medium/High), Product Category (Electronics, Grocery). Classify each as Nominal, Ordinal, Interval, or Ratio and justify.
- (b) **[3 Marks]** A dataset records education level: High School $\rightarrow$ Bachelor's $\rightarrow$ Master's $\rightarrow$ PhD. What would be an accurate characterization of the data type for education level and explain which arithmetic operations (if any) such as computing the mean, are meaningful and which are not.
- (c) **[3 Marks]** Consider a dataset where the values of a numerical attribute for a set of instances are :

10, 12, 13, NaN, 15, 18, NaN, 20 where NaN denotes missing values.

Impute the missing values using mean and median imputation strategies.

Q3

- (a) **[5 Marks]** Consider a function $J(\theta) = (\theta - 6)^2$ which we wish to minimize using gradient descent. Let the initial value for θ be 10, and let the learning rate $\alpha = 0.1$ (assume a fixed learning rate). Perform two iterations of gradient descent and compute updated θ values.

- (b) **[5 Marks]** A 3-class classifier produces the following confusion matrix:

Actual \ Predicted	A	B	C
A	40	5	5
B	8	30	12
C	3	7	35

Compute the following classification measures.

- Accuracy
- Precision, Recall, and F1-score for each class

Q4

- a. **[2 Marks]** Why can't we use the mean square error cost function used in linear regression for logistic regression?
- b. **[3 Marks]** Consider the following dataset and explain with justification whether logistic regression can be used for perfect separation of the two classes.

F1	F2	Class
-1	-1	positive
1	-1	negative
-1	1	negative
1	1	positive

- c. **[5 Marks]** Can you use logistic regression for classification between more than two classes? How many binary classifiers would you need to implement one-vs-one for four classes? Explain how an instance is classified using multiple binary classifiers in the one-vs-one strategy.

Q5

- a. **[5 Marks]** How does L1 and L2 regularization affect bias and variance?
- b. **[5 Marks]** Suppose the true function is: $y=x^3+\epsilon$ where $\epsilon \sim N(0, \sigma^2)$.

Three models are fitted:

Model 1: Linear regression

Model 2: Cubic regression

Model 3: 15th-degree polynomial

Answer the following questions and justify your answer in each case:

- Rank the models in terms of bias (highest to lowest).
- Rank the models in terms of variance (highest to lowest).
- Which model is expected to have the lowest test error? Explain using bias–variance decomposition.

Q6

Consider the following training example dataset, where a1,a2 and a3 refer to attributes and Class refers to the class label (Positive, Negative).

InstanceID	a1	a2	a3	Class
1	T	T	1	+
2	T	T	6	+
3	T	F	5	-
4	F	F	4	+
5	F	T	7	-
6	F	T	3	-
7	F	F	8	-
8	T	F	7	+
9	F	T	5	-

- (a) **[1 Mark]** What is the entropy of this collection of dataset with respect to the class?
- (b) **[4 Marks]** Find the Information gain of a1 and a2 and hence evaluate which amongst a1 and a2 is the better split at the root node level? Assume entropy is used as the node impurity measure.
- (c) **[5 Marks]** Consider attributes a2 and a3 in the table above. Assuming Gini index is used as the node impurity measure evaluate which of the following is the better candidate split?
- Attribute a2 : split based on category values
 - Attribute a3 : split based on threshold = mean value of a3 computed from the data in table above.

Recall that if $p_1, \dots, p_K$ are the estimated probabilities of instances belonging to classes 1, 2, ..., K respectively, the Gini index is given by

$$\text{Gini} = 1 - \sum p_i^2$$